Loss-Aware Synthetic Data Generation:
When Does Privacy Cost Utility, and Who Decides?

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Abstract

Synthetic data is often proposed as a way to share the statistical content of a sensitive dataset without exposing the people in it. In practice the usual workflow is to train a generator, sample from it, and only afterwards measure how much utility was lost and how much privacy risk is left. In this project we argue that this post-hoc approach is problematic from an ethical point of view, because the privacy-utility trade-off ends up being decided by whatever the generator architecture happens to do, and we build an alternative where the trade-off is written into the training objective as explicit weights. We first implement an evaluation framework covering distributional fidelity (MMD, EMD), structural fidelity (correlation preservation), downstream utility (F1 discrepancy) and four empirical privacy indicators (DCR, NNDR, inference risk, disclosure rate). We then extend a tabular variational autoencoder with three differentiable penalty terms (an MMD term, a correlation term and a distance-to-closest-record hinge) and test the model on three public datasets of very different size. We report three main findings. First, regularizing for distributional fidelity also improves privacy when the generator memorizes: on the tabular VAE it increased the distance between synthetic and real records by 17 to 26% on all three datasets while also improving utility, and on a CTGAN that does not memorize the same terms gave no privacy gain. Second, whether an explicit privacy penalty costs utility depends on how densely the real data fills the generator's feature space, and when a cost appears it falls on the least populated parts of each column, that is minority classes, rare categories and the target rate. Third, the standard aggregate utility metrics do not detect these distortions. On a census dataset every fidelity score improved while the sex and income marginals shifted by more than ten percentage points, so a synthetic table can pass every common utility check and still misrepresent who is in it. Since privacy is trivially maximized by generating data that resembles nobody, a privacy score only means something together with a utility floor that includes per-column marginal checks. The ethical consequence is that the populations for whom privacy failures matter most (small, rare, or richly described) are also those for whom protection is most expensive, and that the weighting between privacy and utility cannot be fixed once for all datasets. Our contribution does not settle that judgment; it provides a mechanism that makes it explicit and forces it to be made openly.

1 Introduction

Consider a hospital that holds ten years of admission records, and a research group that wants to train a readmission model on them. Releasing the records directly is not an option. Removing names and identifiers has been known to be insufficient since the 1990s (Sweeney, 2002), and a formal differential-privacy release would require the hospital to choose a privacy budget it has no principled way of picking. Synthetic data sits between these options: train a generative model on the real records, sample a new table of fictional patients, and release that instead.

This proposal has a well-known tension at its centre. The more closely the synthetic table reproduces the real one, the more useful it is, and the more it risks reproducing real individuals. The less closely it reproduces it, the safer and the less useful it is. Every synthetic-data pipeline sits somewhere on this trade-off. What we are interested in with this project is who decides where it sits, and whether they are aware of making that decision.

In current practice, nobody decides and nobody is aware. A generator is trained on a reconstruction or adversarial objective that says nothing about privacy. The samples are then scored with a set of utility and privacy metrics, and if the scores are not acceptable the practitioner changes some hyperparameters and tries again. The trade-off is determined by architectural defaults and tuning habits. It is never stated explicitly and it is never anyone's responsibility.

We think this is an ethical problem as well as a technical one, and we tried to build the alternative: a generator whose training objective contains the privacy-utility trade-off as explicit numerical weights that a stakeholder can read, discuss and set. Concretely, we

1. build an evaluation framework that measures utility loss and privacy risk with eight complementary metrics (§5);
2. extend a tabular variational autoencoder with three differentiable penalty terms, for distribution mismatch, correlation distortion and proximity to real records, so that the trade-off is optimized during training rather than checked afterwards (§6);
3. run the resulting model on three public datasets of very different size and shape, compare it with three baseline generator families, and try to characterize when a privacy penalty costs utility and when it does not (§8).

The results were not quite what we expected. Our initial goal was to *minimize* utility loss under a privacy constraint. What we found instead is that where the loss lands depends on the data. The penalty that protects individuals does so by pushing generated records into the least populated regions of each attribute (minority classes, rare categories, distribution tails), and it does this more aggressively the sparser the data is. On a dense census dataset every aggregate utility metric reported an improvement while the sex and income marginals moved by more than ten percentage points. On a small clinical dataset and on a large high-dimensional one, the same penalty distorted the outcome rate by 23 and 75 points respectively. The relevant variable turned out to be data density, which corresponds closely to the populations for whom privacy matters most, and the standard utility metrics were not able to see the damage. We consider this the main result of the project and we come back to its ethical implications in §9.

1.1 Relation to the original proposal

The project proposal (reproduced in Appendix D) promised three deliverables. We account for each of them here so that the reader can see what was delivered as planned, what changed and why.

Deliverable 1, a computational framework for estimating utility loss. Delivered as `src/evaluation/` and described in §5. The proposal listed MMD, EMD and F1 discrepancy on the utility side and DCR, Overfitting Protection, Inference Risk, CM3 and Disclosure Protection on the privacy side. We implemented MMD, EMD, F1 discrepancy and added correlation distance; on the privacy side we implemented DCR, NNDR (which is the standard implementation of what the proposal called Overfitting Protection), inference risk and disclosure rate. CM3 was not implemented: after reviewing the literature we could not find a definition that was both standard and distinct from the attribute-inference attack we already had, and we preferred fewer metrics we could explain over a longer list. The proposal did not anticipate per-column marginal checks; the results in §8.6 show that these turned out to be the most important part of the framework.

Deliverable 2, a comparative analysis of generator architectures. Delivered in §8.1: CTGAN, TVAE and a Gaussian copula, evaluated on all three datasets with the full metric suite.

The analysis is smaller than the proposal's wording suggests (three families, library defaults, no per-dataset tuning of the baselines) and we say so in §10.

Deliverable 3, a multi-objective loss function. Delivered as `src/generators/loss_aware.py` and described in §6. The proposal listed four penalty terms: distribution mismatch, correlation distortion, downstream performance degradation and privacy leakage. We implemented the first, second and fourth. The third, a penalty on downstream predictive performance, requires training a classifier inside the loss and is not differentiable with respect to the generator in any straightforward way, so it is measured at evaluation time only. The proposal also described the goal as actively *minimizing* utility loss under privacy constraints. As discussed above and in §9, the experiments changed our view of what the interesting question is: not how to minimize the loss, but where it lands, who bears it, and whether the standard metrics can see it.

2 Background and related work

Tabular generative models. Generative adversarial networks and variational autoencoders were adapted to mixed-type tabular data by Xu et al. (2019), whose CTGAN and TVAE are still the most common baselines. Both use a data transformer that one-hot encodes discrete columns and applies *mode-specific normalization* to continuous ones: a Gaussian mixture is fitted per column, and each value is represented as a scalar $\alpha \in [-1, 1]$ within its selected mode plus a one-hot indicator of that mode. The Synthetic Data Vault (Patki et al., 2016) packages these models together with metadata handling and evaluation tools. We build directly on the `ctgan` implementation of TVAE.

Measuring utility. The utility of synthetic data is usually assessed at three levels. Distributional fidelity is measured with two-sample statistics such as Maximum Mean Discrepancy (Gretton et al., 2012) or with per-feature Wasserstein distance (Rubner et al., 2000). Structural fidelity is measured by comparing correlation matrices. Downstream utility is measured with the train-on-synthetic, test-on-real protocol, where a model fitted on synthetic data is evaluated on a real held-out set and compared to one fitted on real data.

Measuring privacy. Empirical privacy indicators for synthetic tables include the Distance to Closest Record (Park et al., 2018), which measures how close synthetic rows are to real ones; the Nearest Neighbour Distance Ratio (Zhao et al., 2022), which detects memorization by comparing the first and second nearest real neighbours; and attribute-inference and membership-inference attacks, which train a model on synthetic data to recover sensitive information about real individuals. Stadler et al. (2022) showed that generators without formal guarantees give inconsistent protection against these attacks, and that "synthetic" should not be read as "anonymous".

Formal privacy. Differential privacy (Dwork, 2006) gives a worst-case guarantee that does not depend on the attacker, and it has been integrated into synthetic-data generators (Xie et al., 2018; Jordon et al., 2019). Our work is complementary to this line: we use empirical indicators because they are interpretable, work with any generator and are directly usable during development, and we are explicit that they are not guarantees (§10).

Loss-level constraints. Adding penalty terms to a generator's objective is common practice for fidelity (for example MMD-GANs) and has also been proposed for fairness. As far as we know, the specific combination we study, namely distributional, structural and distance-based privacy penalties on a TVAE with a self-calibrating margin, has not been characterized across datasets of different density.

3 Ethical framing

We put the ethical argument before the methods because the methods follow from it.

The trade-off is a value judgment, not only a hyperparameter. Any synthetic-data pipeline chooses, implicitly or explicitly, how much fidelity to give up in exchange for how much protection. That choice decides between two groups with different interests: the researchers who will use the synthetic table and who benefit from its fidelity, and the individuals whose records trained the generator and who bear the cost if it leaks. These groups rarely overlap, and neither of them is in the room when a data scientist chooses a learning rate. A design that hides the trade-off inside architectural defaults is not neutral. It decides in favour of whoever benefits from the default, which is usually the researcher.

Privacy by design. Article 25 of the GDPR requires data protection to be built into the design of processing systems rather than added afterwards (European Parliament and Council, 2016; Cavoukian, 2009). A generator that is evaluated for privacy after training satisfies this only if the evaluation actually gates the release, and in practice it often does not. A generator that is *trained* under a privacy constraint satisfies it structurally: the constraint cannot be forgotten, skipped or waived under time pressure, because it is part of what the model optimizes. This is the design principle behind §6.

What the metrics protect, and what they do not. Our privacy indicators measure re-identification and inference risk for individuals in the training set. They do not measure whether the synthetic table preserves or amplifies group-level bias. A faithful synthetic version of a dataset that encodes historical discrimination will encode it as well, and the utility penalties we introduce actively reward that faithfulness. Two of our three datasets contain race, sex and age. We list this as a limitation (§10) rather than addressing it, but it belongs in the framing: being “private” and being “fair” are separate properties, and optimizing one says nothing about the other.

The dual-use concern. A framework that makes synthetic data more useful also makes it more attractive to use in contexts where the real data should not have been collected in the first place. We do not think this is a reason not to do the work, but it is a reason to be careful about what the work shows. The fact that privacy can sometimes be obtained cheaply is not a licence to collect more data.

Consent. Synthetic data is often presented as removing the consent problem, since the records are fictional and nobody’s data is released. Our results complicate this. In the regime where a generator memorizes, synthetic records are essentially slightly perturbed real ones, and the consent problem is hidden rather than removed. Only when the privacy indicators show that the generator has moved away from individual records does the “fictional” framing become honest. That is what our penalties are designed to enforce and our metrics to verify.

3.1 A case study in three readings

The argument above can be made concrete with one example. Consider the Adult census dataset, the loss-aware generator at $\lambda_{\text{priv}} = 10$, $\mu = 1.5$, and the full set of numbers this project produced for it (Tables 5, 2 and 7). Those numbers are established in §8; we use them here, before the methods that produced them, because the ethical argument is easier to follow with real figures than with hypothetical ones. Imagine three people looking at the same page.

The data-protection officer reads the privacy block. The mean distance to the closest real record has more than doubled (1.11 $\rightarrow$ 2.37), the 5th-percentile distance (the most exposed synthetic rows) has grown by a factor of 2.5, and the fraction of synthetic rows within 0.5 standardized units of a real person has dropped from 13% to 2%. She compares this to the stock generators her team has been using: TVAE with 12.6% disclosure, CTGAN with 4.4%. The loss-aware model is the most private neural generator on the table. Her obligation under Article 25 is to build protection into the processing, and here is a generator that does so with a documented weight she can cite. She approves the release.

The researcher reads the utility block. A random forest trained on the synthetic table scores within 0.036 weighted-F1 of one trained on real data, which is at least as good as any baseline and clearly better than the same generator without the privacy term. MMD and correlation distance are no worse than the baseline. Every fidelity metric in the standard toolkit says this synthetic table is a good substitute for the real one, and it is more private than the alternatives. She does not see any trade-off to weigh. She approves the release.

The respondent representative does not start from either block. She asks a simpler question: who is in this synthetic population? The answer, from Table 7, is that the synthetic population has 14 percentage points more of one sex than the real one, its income distribution is 11 points off, and its race and country-of-origin marginals have each moved by roughly one standard deviation, which are the largest shifts of any columns in the table. The generator protected individuals by moving fictional records into the least populated regions of exactly the attributes that define who the population is. If this table is used to study income by sex, race or origin, it will misrepresent all three. She asks the other two whether they knew. They did not, because nothing they read told them.

What the case shows. The three people are not disagreeing about values. They are reading different instruments, and two of those instruments cannot see what the third one sees. The DPO's and the researcher's verdicts are correct for the quantities they measured. The representative's objection is correct for a quantity neither of them measured. If the per-feature analysis of §8.6 had not been run, the release would have been approved unanimously, with a synthetic census that misrepresents its own demographics and that every standard metric certifies as private and useful.

Three points follow from this, and they are the practical content of this report's ethical position. First, the trade-off cannot be judged from aggregate scores alone; the per-column marginals of protected attributes and of the outcome need to be on the same page as DCR and F1, otherwise the page is incomplete. Second, the person who asks "who is in this population?" needs to be in the room, because the other two are unlikely to ask; this is an argument for representation, not only for better metrics. Third, once all three see the same table, they have a real decision to make. They can choose $\lambda_{\text{priv}} = 10$ with a 14-point sex shift, or $\lambda_{\text{priv}} = 2$ with a smaller privacy gain and marginals largely intact, or something in between, and that choice is about whose interests count for how much. The loss-aware generator did not make that decision. It made it possible to see that there was one.

4 Datasets

We use three public tabular datasets from the UCI repository, chosen to vary in domain, size and shape (Table 1). Each loader returns a label-encoded `DataFrame` without missing values and with a binary target, behind a single `DatasetBundle` interface, so that no downstream code depends on which dataset it is handling.

Table 1: Datasets after preprocessing. Row counts are the 80% training split seen by the generator; the remaining 20% is held out for evaluation.

Dataset	Domain	Rows (train)	Columns	Target
Heart Disease (Cleveland) (Detrano et al., 1989)	clinical	237	13	disease present
UCI Adult (Becker and Kohavi, 1996)	socioeconomic	37,000	14	income $>$ 50K
Diabetes 130-US Hospitals (Strack et al., 2014)	clinical	57,000	≈ 40	readmitted $<$ 30 days

Heart Disease is deliberately very small. Generative models tend to memorize on small data, and we wanted the failure mode our privacy metrics are designed to catch to actually be present. **Adult** is the standard benchmark in the fairness and privacy literature and contains demographically sensitive attributes. **Diabetes** contains real hospital admission data with diagnosis codes and medication histories; it is the most ethically sensitive of the three and, with about forty columns, the highest-dimensional. The preprocessing decisions (label encoding rather than one-hot, a 40% missingness threshold for dropping columns on Diabetes, binarization of the Heart Disease severity score) are documented in the repository.

5 Evaluation framework

All metrics take the real training split X_{train}, the real test split X_{test} and the synthetic table $\tilde{X}$, and are pure functions without state. Utility metrics compare $\tilde{X}$ with X_{test} (the unseen distribution); distance-based privacy metrics compare $\tilde{X}$ with X_{train} (the records that could have been memorized). Distances are computed on standardized features after subsampling to at most 2,000 rows.

5.1 Utility

Maximum Mean Discrepancy. With an RBF kernel $k(a,b) = \exp(-\gamma\|a-b\|^2)$, $\gamma = 1$,

$$\mathrm{MMD}^2(\tilde{X}, X) = \mathbb{E}[k(\tilde{x}, \tilde{x}')] + \mathbb{E}[k(x, x')] - 2\,\mathbb{E}[k(\tilde{x}, x)]. \tag{1}$$

The RBF kernel is characteristic, so MMD is zero only if the two distributions are the same.

Earth Mover's Distance. The 1-Wasserstein distance between the empirical marginals, computed per feature and averaged across features. It is reported in raw feature units, which makes it comparable across configurations within a dataset but not across datasets.

Correlation distance. $\|\mathrm{Corr}(X) - \mathrm{Corr}(\tilde{X})\|_F$, the Frobenius norm of the difference between the Pearson correlation matrices.

F1 discrepancy. One random forest is trained on X_{train} and another on $\tilde{X}$. Both are scored on X_{test} with weighted F1, and we report the difference $F1_{\text{real}} - F1_{\text{synth}}$. This is the train-on-synthetic, test-on-real protocol; a positive value means utility was lost.

5.2 Privacy

Distance to Closest Record (DCR). For each synthetic row, the Euclidean distance to its nearest real training row. We report mean, median, minimum and 5th percentile. The tail statistics matter because a small number of near-copies is a serious violation even when the mean looks fine.

Nearest Neighbour Distance Ratio (NNDR). For each synthetic row, d_1/d_2 for its first and second nearest real neighbours. Values close to 0 indicate memorization. Values close to 1 can indicate good generalization, but, as we found out, they also appear when the row is far from everything (§8.3), so NNDR has to be read together with DCR.

Inference risk. A random forest trained on $\tilde{X}$ to predict the sensitive column is scored on X_{test} and compared to a majority-class baseline. In our datasets the sensitive column and the classification target coincide, which makes this metric numerically identical to $F1_{\text{synth}}$: any synthetic data that is useful for predicting heart disease is, by construction, data that lets an attacker infer heart disease. We report it for completeness and discuss the identity in §10.

Disclosure rate. The fraction of synthetic rows within 0.5 standardized units of a real training row, i.e. a direct count of near-copies.

For both families we use the convention that higher utility scores mean more loss and higher privacy scores mean more risk, so that both can be combined directly into a penalty.

6 Loss-aware generation

6.1 Base model

We build on TVAE (Xu et al., 2019), a variational autoencoder (Kingma and Welling, 2014) with the tabular transformer described in §2, trained on the negative evidence lower bound

$$\mathcal{L}_{\text{VAE}} = \underbrace{-\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]}_{\mathcal{L}_{\text{recon}}} + \underbrace{D_{\text{KL}}(q_\phi(z|x) \,\|\, \mathcal{N}(0, I))}_{\mathcal{L}_{\text{KL}}}. \tag{2}$$

We chose TVAE over CTGAN because its training loop is short and has no adversarial component, so it can be subclassed and extended by inserting a single block. Our class overrides only `fit()`. With every penalty weight set to zero it is exactly the stock model, which makes it the fair baseline for all experiments.

6.2 Objective

The loss-aware objective is

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{KL}} + \lambda_{\text{mmd}}\,\mathcal{L}_{\text{mmd}} + \lambda_{\text{corr}}\,\mathcal{L}_{\text{corr}} + \lambda_{\text{priv}}\,\mathcal{L}_{\text{priv}}, \tag{3}$$

with $\lambda_{\text{mmd}}, \lambda_{\text{corr}}, \lambda_{\text{priv}} \geq 0$. The penalties are not computed on reconstructions of the real batch but on a fresh batch decoded from the prior, $\tilde{x}_i = \sigma(\text{decoder}(z_i))$, $z_i \sim \mathcal{N}(0, I)$, because utility and privacy are properties of what the model produces at sampling time. Here σ applies tanh to continuous scalars and either a softmax (soft mode) or a straight-through one-hot (hard mode) to each one-hot span.

Distribution penalty. $\mathcal{L}_{\text{mmd}}$ is the biased MMD^2 estimate from §5 rewritten in `torch`, computed on soft activations.

Correlation penalty. $\mathcal{L}_{\text{corr}} = \|\text{Corr}(\tilde{X}) - \text{Corr}(X)\|_F$ on soft activations, with an ε guard so that constant columns contribute zero instead of NaN. MMD with a fixed bandwidth is mostly sensitive to marginal mismatch; this term puts gradient directly on the second-order structure.

Privacy penalty. This is DCR turned into a gradient. For each generated row, the distance to its nearest real row in the batch is compared with a margin m:

$$d_i = \min_j \|\tilde{x}_i^{\text{hard}} - x_j\|_2, \qquad \mathcal{L}_{\text{priv}} = \frac{1}{B}\sum_{i=1}^{B} \max(0,\, m - d_i). \tag{4}$$

The hinge is zero when every row keeps its distance, and grows linearly as a row gets closer to a real record.

6.3 The same objective on CTGAN

Section 8.7 also tests the three penalties on CTGAN, to check whether the effects found on TVAE depend on the base architecture. CTGAN (Xu et al., 2019) trains a generator G_θ and a discriminator D_ψ with a conditional Wasserstein objective; only the generator side is modified. With $\tilde{x} = \sigma(G_\theta(z, c))$ for a sampled condition vector c, the stock generator loss is

$$\mathcal{L}_G^{\text{stock}} = -\mathbb{E}\big[D_\psi(\tilde{x}, c)\big] + \text{CE}(\hat{c}, c), \tag{5}$$

the critic score plus a cross-entropy term that keeps the generated row consistent with the sampled condition. The loss-aware generator objective adds the same three terms as Equation 3, applied to the same fake batch $\tilde{x}$ and a real batch drawn under the matching condition:

$$\mathcal{L}_G = \mathcal{L}_G^{\text{stock}} + \lambda_{\text{mmd}}\,\mathcal{L}_{\text{mmd}} + \lambda_{\text{corr}}\,\mathcal{L}_{\text{corr}} + \lambda_{\text{priv}}\,\mathcal{L}_{\text{priv}}. \tag{6}$$

$\mathcal{L}_{\text{mmd}}$, $\mathcal{L}_{\text{corr}}$ and $\mathcal{L}_{\text{priv}}$ are defined exactly as before; only the base term they are added to differs (a WGAN critic score instead of a VAE reconstruction and KL term), and the discriminator's own loss and gradient penalty are untouched. Because the critic score has an arbitrary scale, the same numeric λ_{priv} does not carry the same relative weight it did on TVAE; we come back to this in §10. Algorithm 2 in Appendix A gives the full training loop.

6.4 Two things that had to be fixed before the privacy term worked

Our first implementation produced a privacy penalty that was exactly zero during the whole training. Two changes were needed to fix this.

The margin has no natural scale. An absolute m in the transformer's space does not mean the same thing across datasets: every one-hot span can contribute up to $\sqrt{2}$ to a distance, so typical nearest-neighbour distances depend on how many discrete columns the dataset has. We therefore compute, once per fit, the median distance from each real row to its nearest *other* real row, $\tilde{d}_{\text{real}}$, and set $m = \mu\,\tilde{d}_{\text{real}}$. The user-facing parameter μ has a simple interpretation: $\mu = 1$ means that a generated row may not be closer to a real row than real rows are to each other.

Soft one-hots are structurally far from hard ones. Even with the relative margin the hinge stayed at zero. The reason is that real rows are exact one-hots while softmax outputs are not. A generated row that would decode to exactly the same record is still about 0.5 away from it on every discrete span, so the minimum achievable distance was larger than any margin derived from real-to-real distances. For the privacy term we therefore use a deterministic straight-through estimator (Bengio et al., 2013),

$$y = \text{one\_hot}(\arg\max_k \ell_k) - \text{sg}(p) + p, \qquad p = \text{softmax}(\ell), \tag{7}$$

whose forward pass is a true one-hot and whose backward pass is the softmax gradient. We first used Gumbel-softmax (Jang et al., 2017), which crashed intermittently on Apple's MPS backend when its $-\log(\text{Exponential})$ noise produced infinities. Since the noise is not needed for our purpose, we replaced it with the deterministic estimator.

Both fixes are documented in more detail in the repository, together with a third lesson. Our very first comparison showed a large utility collapse under the loss-aware objective, which turned out to be caused by undertraining: with a batch size larger than the dataset, each epoch was a single gradient step. In our experience the first ablation on a new setup tends to measure the training budget rather than the idea being tested.

7 Experimental setup

Every run is described by a `TrainingConfig` with dataset, generator, seed, test fraction and generator keyword arguments. The seed controls the train/test split, the initialization and the sampling. Each run writes a full JSON report (all metrics, per-batch loss components, effective margin, code version) and appends a summary line to a queryable index, so that every number in this report can be reproduced from disk. [TODO: miky: one paragraph on the experiment framework (registry, persistence, CLI)]

Heart Disease runs use `batch_size=32` and 300 epochs; Adult and Diabetes use `batch_size=500` and 100 epochs. The utility weights are fixed at $\lambda_{\text{mmd}} = 1$, $\lambda_{\text{corr}} = 0.5$ whenever they are enabled. We report means across seeds. Wherever a $\pm$ appears it is the half-width of the 95% confidence interval for the mean, computed with the Student-t distribution, $\bar{x} \pm t_{0.975,\,n-1}\, s/\sqrt{n}$. With $n = 3$ seeds the multiplier on the sample standard deviation is about 2.5 and with $n = 5$ about 1.2, so intervals from three seeds are wide by construction. We say explicitly in the text which differences are resolved by these intervals and which are not. Four configurations recur: *baseline* (all $\lambda = 0$), *utility only* ($\lambda_{\text{priv}} = 0$), *privacy only* ($\lambda_{\text{mmd}} = \lambda_{\text{corr}} = 0$) and *both*.

8 Results

8.1 Baseline architectures

Before the loss-aware experiments, we ran three generator families through the evaluation pipeline without modifications: CTGAN and TVAE via the `sdv` library's `CTGANSynthesizer` and `TVAESynthesizer` (which wrap the same `ctgan` package models used directly in §6, with `sdv`'s metadata-driven preprocessing on top), and a Gaussian copula, which is a purely statistical model without a neural network that fits marginals and a correlation matrix.

Table 2: Baseline generators, library defaults, 3 seeds. †CTGAN on Heart is undertrained: `sdv`'s default `batch_size=500` exceeds the 237-row training set, giving one gradient step per epoch.

dataset	generator	MMD	mean EMD	corr. dist	F1 disc	DCR mean	DCR 5th	disclosure
Heart	CTGAN	0.018	5.91	3.47	0.31†	3.30	2.06	0
Heart	TVAE	0.022	1.79	2.85	0.01	1.74	0.83	0.7%
Heart	Gaussian copula	0.018	1.55	2.54	0.05	2.62	1.47	0.1%
Adult	CTGAN	0.0019	1274	0.79	0.053	1.68	0.53	4.4%
Adult	TVAE	0.0024	3878	0.97	0.058	1.18	0.32	12.6%
Adult	Gaussian copula	0.0024	2310	1.26	0.145	3.08	1.20	0.03%
Diabetes	CTGAN	0.0015	2.98	2.54	0.004	5.78	2.26	0
Diabetes	TVAE	0.0017	3.84	4.21	0.004	2.58	1.51	0
Diabetes	Gaussian copula	0.0015	1.44	2.77	0.004	4.52	2.85	0

We make four observations from Table 2.

TVAE is the least private architecture on every dataset. It has the lowest DCR and the highest disclosure rate in each block (12.6% on Adult, against 4.4% for CTGAN and 0.03% for the copula), while being the most useful on Heart and Adult. The loss-aware experiments from §8.3 onward therefore start from the generator with the most room for privacy improvement, and the size of the "free lunch" effect we report there may partly be a property of TVAE.

The Gaussian copula is competitive without a neural network. It has the lowest EMD on Heart and Diabetes, the lowest correlation distance on Heart, and is always more private than TVAE. On Adult it pays for this with the worst F1 discrepancy by a factor of three; on both clinical datasets it is the most balanced baseline. Lower capacity leads to less memorization, which leads to more privacy. This is the same density argument as in §8.5, seen from the model side rather than the data side.

CTGAN trades utility for privacy. It is more private than TVAE everywhere and less useful on Adult. On Diabetes its DCR mean (5.8) is far above its median (3.8), which indicates a long tail of samples far from any real record. This is the off-manifold minority that appears again in §8.3. Its Heart row is not usable as reported; with a tuned training budget (§8.7) CTGAN reaches F1 discrepancy 0.01 on Heart while keeping DCR 2.6.

The stock model matches our baseline. `sdv`'s TVAE is within seed noise of our $\lambda = 0$ subclass on all three datasets (Heart: DCR 1.74 vs 1.72; Adult: 1.18 vs 1.11; Diabetes: 2.58 vs 2.48). This supports the claim in §6 that the loss-aware class with zero weights is TVAE, rather than just asserting it.

We come back to this table in §9 to compare the loss-aware model against it.

8.2 Heart Disease: ablation

Table 3: Heart Disease ablation, $\mu = 1$, 3–4 seeds. $\pm$ is the 95% CI half-width. Privacy-only rows predate the fixes of §6.4 and should be read as "hinge inactive".

configuration	MMD	corr. dist	F1 disc	DCR mean	DCR 5th	disclosure
baseline	0.0193	2.66 ± 0.55	-0.03 ± 0.09	1.72 ± 0.13	0.74 ± 0.17	1.0%
utility only	0.0178	1.91 ± 0.74	-0.01 ± 0.06	2.02 ± 0.10	1.01 ± 0.05	0.4%
privacy only ($\lambda_{\text{priv}} = 1$)	0.0192	2.37 ± 0.18	-0.00 ± 0.04	1.76 ± 0.06	0.80 ± 0.04	0.9%
both ($\lambda_{\text{priv}} = 1$)	0.0179	1.94 ± 0.89	-0.03 ± 0.00	2.02 ± 0.01	0.98 ± 0.09	0.4%

Three things stand out in Table 3. The two terms that are in the loss (MMD and correlation) improve on average, while EMD, which is not in the loss, does not; the correlation-distance intervals overlap, though, so with three seeds that improvement is suggestive rather than established. F1 discrepancy is indistinguishable from zero in every row, so with a sufficient training budget the synthetic data is as useful as real data for this task. Finally, the utility terms improved privacy, and this one is resolved: on their own they raised mean DCR from 1.72 ± 0.13 to 2.02 ± 0.10, raised the 5th-percentile DCR from 0.74 ± 0.17 to 1.01 ± 0.05 and halved the disclosure rate. We initially attributed this to reduced memorization on a small dataset. §8.4 and §8.5 show that the effect does not shrink with dataset size, so the mechanism appears to be more general.

8.3 Heart Disease: finding the trade-off

With the fixes in place, the hinge behaves as one would expect from a hinge. At $\lambda_{\text{priv}} = 1$ its effect grows with μ and is concentrated in the tail (5th-percentile DCR $0.74 \to 0.78 \to 0.80 \to 0.83$ for

$\mu = 0, 0.5, 1, 1.5$), while the mean DCR barely moves. It is also weak: reconstruction and KL are of order 10, the hinge of order 0.1. We therefore swept the weight.

Table 4: Heart Disease weight sweep, "both" configuration. $\pm$ is the 95% CI half-width; rows for $\lambda_{\text{priv}} \in \{2, 5, 10\}$ at $\mu = 1.5$ have 5 seeds, the others 3.

λ_{priv}	μ	F1 disc	DCR mean	DCR 5th	mean EMD	corr. dist	disclosure
0 (baseline)	–	-0.03 ± 0.09	1.72 ± 0.13	0.74 ± 0.17	1.36 ± 0.44	2.66 ± 0.55	1.0%
1	1.5	-0.02 ± 0.07	2.04 ± 0.03	1.04 ± 0.09	1.21 ± 0.07	1.92 ± 0.98	0.2%
2	1.5	-0.01 ± 0.08	2.06 ± 0.05	0.98 ± 0.05	1.37 ± 0.39	2.00 ± 0.19	0.3%
5	1.5	-0.02 ± 0.05	2.13 ± 0.07	1.08 ± 0.04	1.47 ± 0.58	2.12 ± 0.31	0.3%
10	1.5	0.07 ± 0.08	2.73 ± 0.23	1.45 ± 0.18	2.46 ± 0.62	2.28 ± 0.32	0.04%
10	2.0	0.22 ± 0.57	6.23 ± 1.68	3.23 ± 0.70	23.6 ± 21.2	3.90 ± 1.24	0
50	1.5	0.08 ± 0.11	6.27 ± 0.98	3.64 ± 1.08	30.5 ± 19.7	3.23 ± 1.04	0
50	2.0	0.21 ± 0.51	6.36 ± 1.34	5.12 ± 1.07	24.6 ± 7.0	3.48 ± 0.91	0

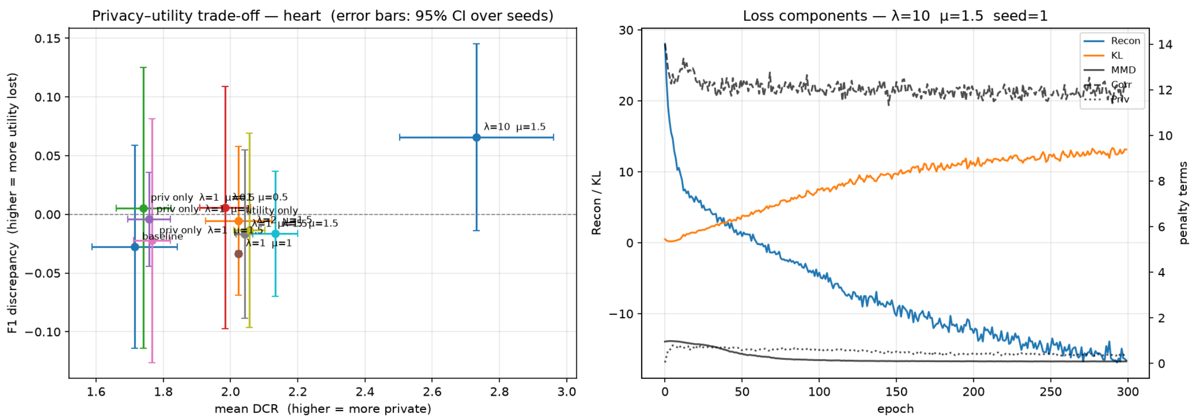


Figure 1: Left: F1 discrepancy against mean DCR on Heart Disease, one point per configuration, error bars are 95% confidence intervals over seeds, collapse regime excluded. Right: loss components over training for one $\lambda_{\text{priv}} = 10$, $\mu = 1.5$ run.

Table 4 shows three regimes.

Regime I, free lunch ($\lambda_{\text{priv}} \le 5$). Privacy improves and nothing is paid for it. F1 discrepancy is indistinguishable from zero and the distributional metrics move by a few percent, within their intervals. The DCR gain over baseline is resolved already at $\lambda_{\text{priv}} = 1$ (2.04 ± 0.03 against 1.72 ± 0.13).

Regime II, trade-off ($\lambda_{\text{priv}} = 10$, $\mu = 1.5$). Mean DCR increases by 33% (2.73 ± 0.23, clearly separated from $\lambda_{\text{priv}} = 5$), the worst-case tail by 47%, and disclosure is zero. A cost appears at the same time, but with five seeds only part of it is resolved. Mean EMD rises to 2.46 ± 0.62 against a baseline of 1.36 ± 0.44, which is a real difference. F1 discrepancy is 0.07 ± 0.08, an interval that includes zero, so the downstream-utility cost is an estimate we cannot distinguish from no cost. Correlation distance (2.28 ± 0.32) is not separated from the baseline either. This is the operating point where privacy starts to have a price, and the price is visible in the marginals (§8.6) more clearly than in F1.

Regime III, collapse ($\mu = 2$ or $\lambda_{\text{priv}} = 50$). Mean DCR saturates around 6.3 regardless of the exact settings, EMD grows by a factor of about twenty, and in one run synthetic F1 is exactly equal to the majority-class baseline, meaning the generated target column collapsed to a single value. The confidence intervals in these rows are very wide (23.6 ± 21.2 for EMD, 0.22 ± 0.57 for F1 discrepancy); the runs disagree with each other about *how* the collapse happens, which is itself a sign that the generator has left the data manifold. Privacy is "perfect" because the samples do not resemble anyone.

Two methodological findings come out of Regime III. The first is that *fixed-bandwidth MMD saturates*: it moved from 0.018 to 0.032 while EMD moved from 1 to 25. Once the two distributions stop overlapping, every cross-kernel term is close to zero and MMD^2 flattens to a constant. As an evaluation metric it under-reports gross failure; as a loss term its gradient vanishes exactly when it is needed most, which is why it could not restrain the hinge at $\lambda_{\text{priv}} = 50$. The second is that *NNDR inverts off the manifold*: it rose to 0.95 in the collapse regime, which the usual reading would interpret as excellent generalization. That reading is wrong here, because a row that is far from all real records has equidistant neighbours for trivial reasons. NNDR is only interpretable together with DCR.

A third observation concerns the measurement itself. The 95% interval for F1 discrepancy is about ± 0.08 at *every* λ_{priv} in Table 4, including where the mean is zero. This is the resolution limit of a 60-row test split, not a property of the generator, and it is why the $\lambda_{\text{priv}} = 10$ cost cannot be separated from zero. The distributional metrics do not have this limit and show the cost. On small datasets, distributional metrics can resolve the trade-off while downstream predictive metrics cannot.

8.4 Adult: the same penalty appears to be free

Table 5: UCI Adult, 3 seeds. $\pm$ is the 95% CI half-width.

configuration	F1 disc	MMD	corr. dist	DCR mean	DCR 5th	disclosure
baseline	0.042 ± 0.031	0.0035	0.87 ± 0.22	1.11 ± 0.07	0.31 ± 0.07	13.4%
utility only	0.048 ± 0.004	0.0021	0.69 ± 0.09	1.36 ± 0.19	0.41 ± 0.05	8.5%
both ($\lambda_{\text{priv}} = 10$, $\mu = 1.5$)	$\mathbf{0.036 \pm 0.005}$	0.0022	0.73 ± 0.18	$\mathbf{2.37 \pm 0.23}$	$\mathbf{0.78 \pm 0.11}$	**1.8%**

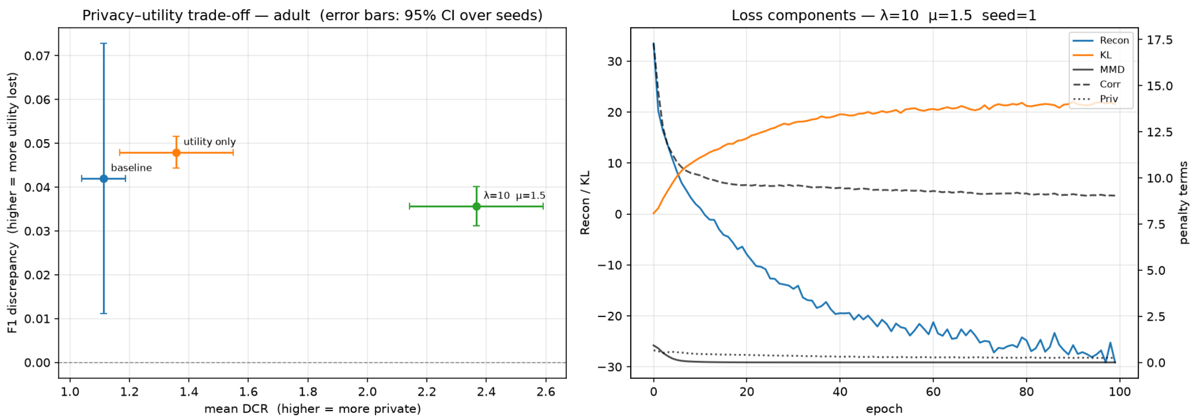


Figure 2: Adult: trade-off (left, error bars are 95% CIs) and loss components (right). Note that $\mathcal{L}_{\text{priv}}$ plateaus at a non-zero value instead of decaying: the model carries a constant cost without visible strain.

On Adult, the configuration that was the knee of the Heart curve improves everything at once by the aggregate metrics (Table 5). Mean DCR more than doubles (2.37 ± 0.23 against 1.11 ± 0.07), the 5th-percentile DCR grows by a factor of 2.5, disclosure drops from 13% to 2%, and F1 discrepancy does not get worse: 0.036 ± 0.005 against a baseline of 0.042 ± 0.031. The two intervals overlap, so we cannot claim an improvement over the baseline, but the loss-aware value is clearly below the utility-only one (0.048 ± 0.004). MMD and correlation distance are both at or better than baseline, the latter within overlapping intervals. F1 is now a much more precise instrument (± 0.005 against ± 0.08 on Heart, since the test split has 9,600 rows), which confirms the resolution-limit explanation by contrast; the baseline's wider interval (± 0.031) comes from one seed and is the exception.

The utility-only row is interesting for a different reason. It improved MMD by 40% and correlation distance by 20% while F1 discrepancy moved slightly in the wrong direction. The F1 intervals overlap with the baseline, so we do not consider this firm, but it is the second time two "utility" metrics point in different directions: matching moments does not guarantee that the decision boundary a classifier needs is preserved. We read this as an argument against reporting a single utility number.

8.5 Diabetes: density is not row count

Table 6: Diabetes 130-US Hospitals, 3 seeds per row; $\mu = 1.5$ and $\lambda_{\text{mmd}} = 1$, $\lambda_{\text{corr}} = 0.5$ wherever $\lambda_{\text{priv}} > 0$. $\pm$ is the 95% CI half-width.

λ_{priv}	F1 disc	F1 synth	MMD	mean EMD	corr. dist	DCR mean	DCR 5th
0 (baseline)	0.004 ± 0.001	0.838	0.0019	3.85 ± 0.44	4.37 ± 0.43	2.48 ± 0.05	1.48 ± 0.05
0 (utility only)	0.004 ± 0.001	0.838	0.0015	2.23 ± 3.16	2.76 ± 0.45	3.13 ± 0.10	1.79 ± 0.13
2	0.004 ± 0.002	0.838	0.0015	$\mathbf{1.66 \pm 1.67}$	$\mathbf{2.65 \pm 0.45}$	3.35 ± 0.10	1.98 ± 0.13
5	0.048 ± 0.042	0.794	0.0015	3.52 ± 1.16	3.29 ± 0.52	7.32 ± 1.25	3.86 ± 0.36
10	0.069 ± 0.203	0.773	0.0016	5.58 ± 2.71	3.05 ± 0.44	10.6 ± 8.2	9.13 ± 9.43

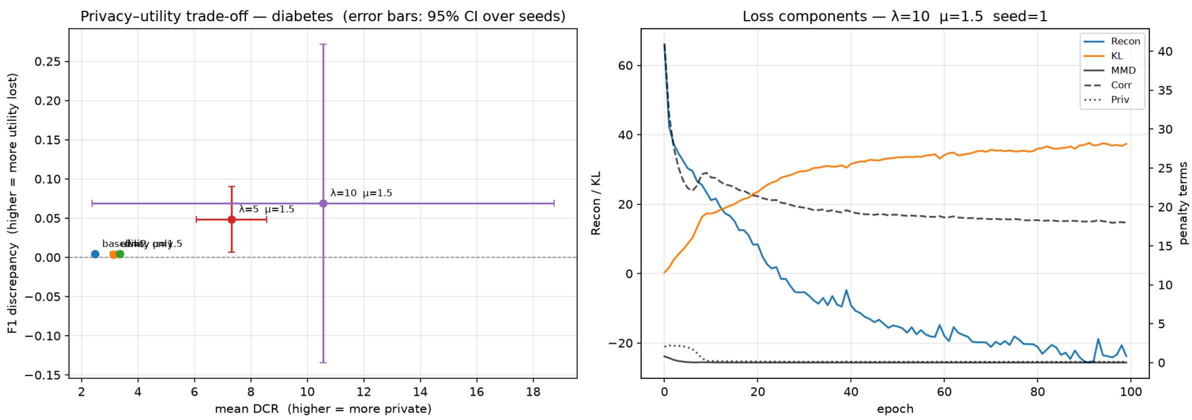


Figure 3: Diabetes: trade-off (left, error bars are 95% CIs) and loss components for a $\lambda_{\text{priv}} = 5$ run (right). Note the discontinuity in DCR between $\lambda_{\text{priv}} = 2$ and 5: the transition to the collapse regime is a jump rather than a slope.

Diabetes has twice as many rows as Adult and still pays for privacy (Table 6, Figure 3). The knee is between $\lambda_{\text{priv}} = 2$ and 5, earlier than on Heart, and it is sharp: mean DCR more than doubles in that single step ($3.35 \pm 0.10 \to 7.32 \pm 1.25$), while the same interval on Heart moved it from 2.06 to 2.13. The generator seems to stay on the manifold until it cannot, and then leaves it. NNDR jumping from 0.91 to 0.96 at the same point is the off-manifold signature discussed in §8.3. At $\lambda_{\text{priv}} = 10$ the interval on mean DCR is ± 8.2: one seed reached 14.3, the other two about 8.7. That spread is itself a symptom of collapse, and we treat $\lambda_{\text{priv}} = 5$ rather than 10 as the reliable description of what happens past the knee. Below the knee, $\lambda_{\text{priv}} = 2$ is the best configuration in the table: lowest EMD and correlation distance of any row (the correlation improvement over baseline, 2.65 ± 0.45 against 4.37 ± 0.43, is well resolved; the EMD one is not), DCR +35% over baseline, F1 unchanged.

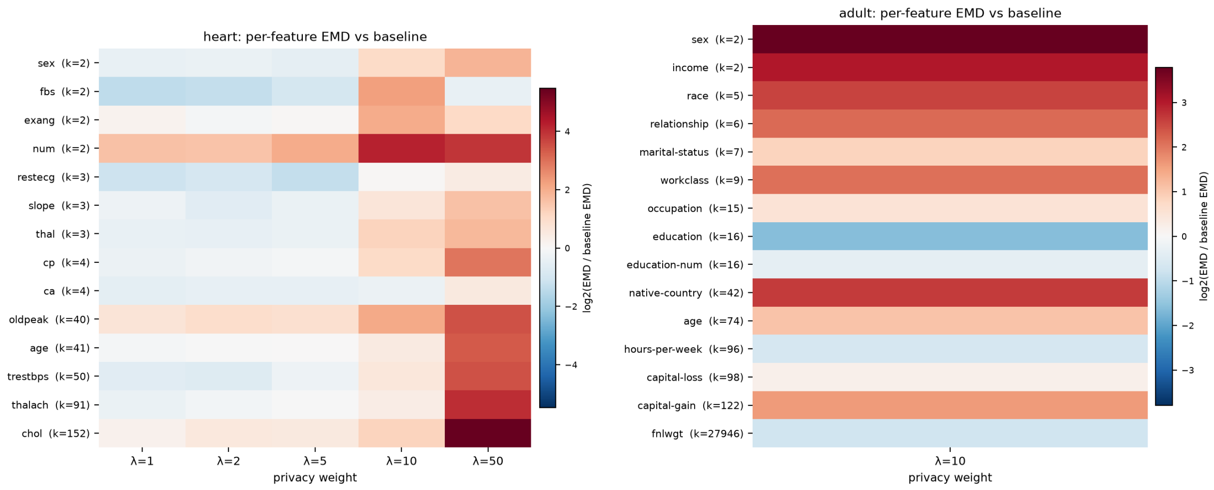
Two more points. First, F1 discrepancy is degenerate on this dataset for a third, different reason: the target is roughly 89/11 imbalanced and weighted F1 sits on a ceiling where the synthetic-trained classifier predicts the majority class ($F1_{\text{synth}} = 0.838$ = majority baseline below the knee) and the real-trained one barely does better (0.842). Above the knee $F1_{\text{synth}}$ drops *below* the majority baseline, which means the synthetic label marginal itself has been distorted. Here F1 discrepancy measures corruption of the target distribution rather than loss of predictive signal, and at $\lambda_{\text{priv}} = 5$ its interval (0.048 ± 0.042) only just excludes zero. Heart's F1 was limited by noise, Adult's was informative, and Diabetes's is limited by class imbalance; positive-class F1 or AUROC would have been a better choice. Second, the utility terms improved privacy again (DCR 3.13 ± 0.10 against 2.48 ± 0.05), together with a resolved 37% reduction in correlation distance. Across three datasets spanning two orders of magnitude in size, the direction is the same and the DCR difference is resolved on all three.

8.6 Where the cost lands: per-feature analysis

Every run stores the EMD of each column separately. Averaging across seeds and comparing each λ_{priv} with the plain baseline tells us *which* columns pay for privacy. For a binary label-encoded column, EMD equals the absolute shift in proportion, so those entries can be read directly as percentage points. Figure 4 shows the heatmaps for Heart and Adult, Figure 5 the one for Diabetes, and Table 7 lists the columns that moved most.

Table 7: Per-feature EMD, mean over seeds. k = number of distinct values. For binary columns EMD is the absolute difference in proportion. Adult rows are the top of the ranking by standardized change (EMD change divided by the real column's standard deviation); the full ranking's top six are native-country, race, workclass, sex, income, relationship, all categorical, four of them protected or sensitive attributes and one the target.

dataset	config	column	baseline	EMD	reading (deviation from real; baseline in parentheses)
Adult	$\lambda_{\text{priv}} = 10$	native-country (k=42)	1.4	9.1	+1.0 sd; largest standardized shift
Adult	$\lambda_{\text{priv}} = 10$	race (k=5)	0.16	0.91	+0.9 sd
Adult	$\lambda_{\text{priv}} = 10$	sex (k=2)	0.010	0.144	sex ratio off by 14 pp (1 pp)
Adult	$\lambda_{\text{priv}} = 10$	income (k=2, target)	0.013	0.111	positive rate off by 11 pp (1 pp)
Adult	$\lambda_{\text{priv}} = 10$	fnlwgt, education, hours/wk	—	—	*improved* (×0.3–0.6)
Heart	$\lambda_{\text{priv}} = 10$	num (k=2, target)	0.013	0.229	disease prevalence off by 23 pp (1 pp)
Heart	$\lambda_{\text{priv}} = 10$	sex (k=2)	0.057	0.118	sex ratio off by 12 pp (6 pp)
Heart	$\lambda_{\text{priv}} = 50$	chol	6.3	280	mean cholesterol off by 280 mg/dL
Heart	$\lambda_{\text{priv}} = 50$	trestbps	5.3	59	resting BP off by 59 mmHg
Heart	$\lambda_{\text{priv}} = 50$	thalach	3.7	62	max heart rate off by 62 bpm
Heart	$\lambda_{\text{priv}} = 50$	age	2.0	20.6	age off by 20 years
Diabetes	$\lambda_{\text{priv}} = 2$	diag_1 (k=656)	46.5	8.5	*improved* 5×
Diabetes	$\lambda_{\text{priv}} = 2$	gender (k=2)	0.125	0.012	*improved* 10×
Diabetes	$\lambda_{\text{priv}} = 10$	readmitted (k=2, target)	0.11	0.75	readmission rate off by 75 pp (11 pp)
Diabetes	$\lambda_{\text{priv}} = 10$	diabetesMed (k=2)	0.016	0.39	medication flag off by 39 pp (2 pp)
Diabetes	$\lambda_{\text{priv}} = 10$	acarbose (k=4)	0.003	0.33	rare drug (∼0.3% real) in ∼1/3 of rows



(a) Heart Disease **(b)** UCI Adult

Figure 4: Per-feature EMD relative to the plain baseline, $\log_2$ ratio: red = worse than baseline, blue = better. Rows sorted by cardinality k (in parentheses); columns are λ_{priv}. On Heart the continuous clinical measurements dominate the collapse column; on Adult the categorical and sensitive columns turn red while the continuous ones turn blue.

The Adult "free lunch" was not free. This is the most important block of Table 7. At $\lambda_{\text{priv}} = 10$ on Adult every aggregate metric improved (Table 5), and we called it the free-lunch region. If we rank the columns by standardized change, the six that moved most are native-country, race, workclass, sex, income and relationship. All of them are categorical, four are protected or sensitive attributes and one is the target, while the continuous columns are at the bottom of the ranking or actually *improved*. The sex ratio ended up 14 points away from the real one and the income rate 11 points. The cost fell exactly on the columns a fairness audit would look at first, and no aggregate metric registered it. Weighted F1 is scored on real test rows, and a classifier trained on a table with too many rows of one sex still classifies real rows well. Pearson correlation is almost invariant to a marginal shift. A fixed-bandwidth MMD in fourteen dimensions hardly registers a single binary coordinate moving. In other words, the verdict "free" came from instruments that are structurally unable to see the ethically relevant distortion. §8.4 is correct for the metrics it reports; we withdraw its interpretation.

What breaks is the sparsest region of each column. Our cardinality test gave inconsistent results across datasets ($r = +0.76$ on Heart, -0.56 on Adult, undefined on Diabetes), and we think this is because cardinality is the wrong variable. What degrades is the minority value of binary columns (sex, the target), rare categories (acarbose, prescribed to 0.3% of real patients and present in a third of synthetic rows), and the tails of continuous columns (cholesterol off by 280 mg/dL in the collapse regime). High-cardinality columns sometimes *improve*. The hinge appears to escape real records by moving probability mass into whatever region of each column is least populated. This is the density account of §8.5 at the level of individual features, and it also suggests a mechanism. It also means that the collapse regime does not just produce implausible records; it produces records that systematically over-represent rare categories and minority classes.

Regime II costs are target-rate shifts. On Heart at $\lambda_{\text{priv}} = 10$ the "0.07 of F1" we reported in §8.3 corresponds to a 23-point shift in disease prevalence. On Diabetes at $\lambda_{\text{priv}} \geq 5$ the target shifts by up to 75 points, which inverts the synthetic majority class and explains why synthetic F1 fell below the majority baseline. On every dataset, the knee of the trade-off is the point where the generator starts to misrepresent how common the outcome is.

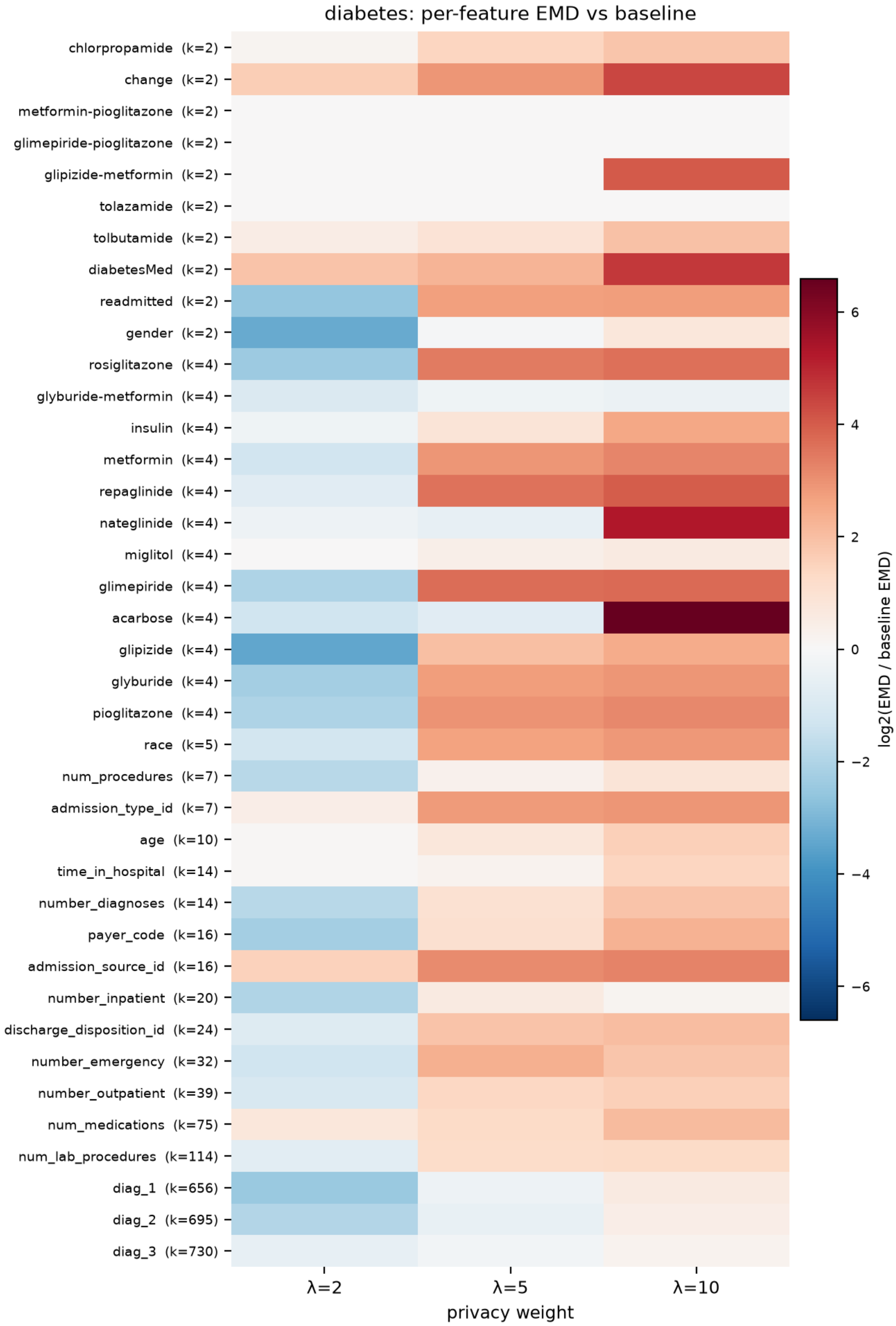


Figure 5: Diabetes 130-US Hospitals: per-feature EMD relative to baseline, same encoding as Figure 4; constant columns omitted. The $\lambda_{\text{priv}} = 2$ column is almost entirely blue; the $\lambda_{\text{priv}} \geq 5$ columns are dominated by the binary medication flags and the readmission target.

Low λ_{priv} is a real sweet spot. At $\lambda_{\text{priv}} = 2$ on Diabetes almost every column improves, the 656-value diagnosis code by a factor of five and gender by a factor of ten. At $\lambda_{\text{priv}} \leq 5$ on Heart most ratios are below one. The free-lunch region of §8.3 exists, but it is narrower than the aggregate metrics suggested, and on Adult it does not extend to $\lambda_{\text{priv}} = 10$.

8.7 Does the effect depend on the generator? Loss-aware CTGAN on Heart

The baseline comparison (§8.1) showed TVAE to be the least private of the three architectures, which raised the possibility that the privacy gain from the utility terms was a TVAE property rather than a general one. To test this we added the same three penalty terms to the generator objective of CTGAN (the discriminator is unchanged) and ran the three main configurations on Heart Disease with `batch_size=50` and 2,000 epochs, about 8,000 generator steps, which is where the stock model converges on this dataset. The implementation is in `src/generators/loss_aware_ctgan.py`.

Table 8: Loss-aware CTGAN on Heart Disease, 3 seeds, $\mu = 1.5$. $\pm$ is the 95% CI half-width. TVAE rows from Tables 3 and 4 for comparison.

generator	configuration	F1 disc	mean EMD	corr. dist	DCR mean	DCR 5th	NNDR
CTGAN	baseline	0.012 ± 0.082	2.57 ± 0.46	2.14 ± 0.78	2.59 ± 0.25	1.37	0.89
CTGAN	utility only	0.005 ± 0.048	3.14 ± 0.75	1.88 ± 0.45	2.54 ± 0.08	1.34	0.88
CTGAN	both ($\lambda_{\text{priv}} = 10$)	0.044 ± 0.068	17.3 ± 4.1	2.38 ± 0.97	5.03 ± 0.42	3.61	0.94
TVAE	baseline	-0.03 ± 0.09	1.36 ± 0.44	2.66 ± 0.55	1.72 ± 0.13	0.74	0.83
TVAE	utility only	-0.01 ± 0.06	1.25 ± 0.54	1.91 ± 0.74	2.02 ± 0.10	1.01	0.85
TVAE	both ($\lambda_{\text{priv}} = 10$)	0.07 ± 0.08	2.46 ± 0.62	2.28 ± 0.32	2.73 ± 0.23	1.45	0.88

The utility terms do not improve privacy on CTGAN. Mean DCR goes from 2.59 to 2.54, with overlapping intervals. Correlation distance improves, as it should since the term is in the loss, and MMD moves slightly, but there is no privacy gain. This settles the question raised in §8.1: the effect is specific to TVAE, and the reason is visible in the table. A properly trained CTGAN starts at DCR 2.59, which is already above the level the utility terms lifted TVAE to (2.02). TVAE at 1.72 was partially memorizing its training rows; the utility terms corrected that; CTGAN did not have the problem in the first place. The memorization hypothesis we set aside in §8.4 therefore returns in a more precise form: fidelity regularization acts as a privacy mechanism for generators that memorize, and its effect is bounded by how far below the non-memorizing level the baseline sits. On TVAE it was 17–26% on all three datasets; on CTGAN it is zero.

The hinge works on CTGAN, and lands harder. At $\lambda_{\text{priv}} = 10$ mean DCR nearly doubles ($2.59 \to 5.03$), the 5th percentile goes from 1.37 to 3.61, and NNDR reaches 0.94, the off-manifold signature of §8.3. Mean EMD rises seven-fold while F1 discrepancy barely moves ($0.012 \to 0.044$, interval including zero). This is the collapse pattern that TVAE showed at $\lambda_{\text{priv}} = 10$, $\mu = 2$, arriving one step earlier, and it is consistent with the hinge-scale confound in §10: the WGAN generator loss has a different scale from a VAE reconstruction loss, so the same numeric λ_{priv} is a stronger penalty here. A sweep at $\lambda_{\text{priv}} \in \{2, 5\}$ would locate CTGAN's knee; we did not have time to run it.

A properly trained CTGAN is a good Heart baseline. With a batch size that gives several steps per epoch and a budget of 8,000 steps, CTGAN reaches F1 discrepancy 0.01 and correlation distance 2.14 (better than TVAE's 2.66) at DCR 2.59 (much more private than TVAE's 1.72), at the cost of a worse EMD (2.57 against 1.36). The † row of Table 2 was an artefact of the library default batch size and should be read against this one.

8.8 Synthesis

Table 9: The three datasets on one axis, ordered by where the trade-off begins.

dataset	rows	columns	density	privacy penalty at $\lambda_{\text{priv}} = 10$	knee (λ_{priv})
Adult	37,000	14	dense	aggregate metrics: free; protected-attribute marginals: 14 pp sex shift	> 10 (aggr.)
Heart	237	13	sparse (few rows)	costs utility	5–10
Diabetes	57,000	≈ 40	sparse (many dims)	costs utility	2–5

Table 9 summarizes the central empirical claim of the project. The hinge pushes each generated row away from its nearest real row. On a dense manifold the row ends up next to another plausible but fictional record and the distribution is preserved. On a sparse one the nearest empty space is off the manifold, the row becomes an implausible record and the distribution degrades. A dataset can be sparse because it has few rows (Heart) or because it has many dimensions (Diabetes: volume grows exponentially with dimensionality, and 57,000 rows in forty dimensions are much thinner than 37,000 in fourteen). What sets the price of a privacy penalty seems to be how densely the records fill the space the generator works in, rather than the number of records. The ordering of the knees is consistent with this account. It is also consistent with the hinge-scale confound discussed in §10, and we are not able to separate the two with our runs.

9 Discussion

Privacy versus utility is an incomplete framing. Regime I shows that on every dataset a term designed only to improve distributional fidelity also improved every privacy indicator on TVAE, by 17 to 26% in mean DCR. Regularizing towards population-level statistics spreads generated samples across the distribution instead of clustering them on training records, and better generalization turns out to be better privacy. The loss-aware CTGAN experiment (§8.7) shows the limit of this: on a generator that does not memorize to begin with, the same terms give no privacy gain. The mechanism is a correction of memorization, and it does as much good as there is memorization to correct. The common picture of two quantities on opposite ends of a dial still misses this region, which exists for the most widely used tabular VAE on all three datasets we tried. The per-feature analysis makes the region narrower (on Adult it does not reach $\lambda_{\text{priv}} = 10$) but does not remove it: at $\lambda_{\text{priv}} = 2$ on Diabetes the diagnosis codes, the gender marginal and the privacy indicators all improved together.

Aggregate utility metrics are blind to the distortions that matter most. The clearest result of this study is a negative one. Every aggregate utility metric we used (MMD, correlation distance, downstream F1) reported the Adult $\lambda_{\text{priv}} = 10$ configuration as an improvement, while the sex ratio moved by 14 points and the income rate by 11. These are the columns a fairness audit would check first, and the standard utility toolkit could not see them move. A synthetic dataset can pass every fidelity check in the literature and still misrepresent who is in it. Per-column marginal checks on protected attributes and on the target are therefore not optional additions to a utility evaluation. Without them, the evaluation does not measure what it claims to measure.

A privacy score without a utility floor is meaningless. Regime III shows that every privacy indicator we use can be trivially maximized by a generator that produces noise. This is obvious once stated but is routinely ignored in practice: “our synthetic data has DCR of x”

is not a statement about privacy unless it comes with a statement about what the data is still good for. Any evaluation protocol, and any regulation, that scores privacy without also scoring utility can be gamed by destroying the data.

The trade-off is a decision, and this is where it lives. Regime II is the region where privacy has a price, and the price is now a number with an uncertainty attached. On Heart Disease, a 33% increase in the distance between synthetic and real patients was bought with a downstream F1 cost of 0.07 ± 0.08, an interval that includes zero, and with a measurably distorted marginal distribution. A data-protection officer, a clinical researcher and a patient representative would weigh that offer differently, and one of the things they would have to weigh is how uncertain the cost is. The contribution of the loss-aware formulation is not that it resolves the disagreement, but that it produces a concrete offer for them to disagree about: four numbers, $\lambda_{\text{mmd}}, \lambda_{\text{corr}}, \lambda_{\text{priv}}, \mu$, that can be documented, argued over and set by whoever has the standing to set them. A stock generator embeds the same decision in defaults that no stakeholder ever sees.

The price falls on those who can least afford it. Table 9 has an ethical reading that we did not anticipate, and that we consider the most important point of this report. The same penalty was free on a dense census dataset, at least by the aggregate metrics, and costly on both sparse ones. Sparsity comes from having few records or from describing each record richly. Small populations, rare conditions, detailed clinical histories and under-represented groups are exactly the settings in which a privacy failure does the most harm, and they are also the settings in which protection is most expensive in terms of utility. The per-feature analysis sharpens this to the level of single attributes. The hinge protects individuals by moving generated mass into the least populated regions of every column, which means it systematically inflates minority classes and rare categories. A synthetic table made private in this way over-represents the very groups whose sparsity made them vulnerable in the first place, and it misstates how common the outcome is. Any policy that fixes a single privacy-utility weighting for all datasets will therefore either under-protect the sparse population or over-pay on the dense one, and where a payment is made, it is made in the demographic marginals. The weighting has to be set per dataset, the density of the data is one of the things whoever sets it needs to see, and the check on what was paid has to be done column by column.

Against the baselines. Table 2 lets us position the loss-aware model. On Adult at $\lambda_{\text{priv}} = 10$ it has F1 discrepancy 0.036 ± 0.005, against 0.053, 0.058 and 0.145 for CTGAN, TVAE and the copula, and mean DCR 2.37 ± 0.23 against 1.68, 1.18 and 3.08. It is therefore at least as useful as every baseline (the CTGAN interval, ± 0.028 over three seeds, overlaps it) and clearly more private than both neural ones. On Heart at the $\lambda_{\text{priv}} \leq 5$ sweet spot it matches the copula's privacy with somewhat better correlation preservation. On Diabetes at $\lambda_{\text{priv}} = 2$ the copula is still more private and the two are close on fidelity. A trained penalty does not make the simple statistical model obsolete, but it does make the neural one competitive with it on privacy while keeping the neural one's utility.

Metrics disagree, and that is useful information. Four times in this study one metric moved while another did not: MMD saturated while EMD exploded; NNDR rose while DCR indicated the samples had left the manifold; MMD and correlation improved while F1 drifted; and every aggregate utility metric improved while the sex and income marginals shifted by more than ten points. Each of these disagreements pointed to a real property of the data or of the metric. A single utility number and a single privacy number would have hidden all four.

10 Limitations

- **Few seeds.** Most configurations have three seeds, the Heart transition region five. The 95% intervals are wide as a result, and several of the utility-cost differences we describe are not statistically resolved: the Heart $\lambda_{\text{priv}} = 10$ F1 cost includes zero, the Adult loss-aware F1 improvement over baseline is not separated, and the Diabetes $\lambda_{\text{priv}} = 10$ DCR has an interval of ± 8. The privacy differences (DCR, disclosure) are resolved throughout, and the marginal shifts in §8.6 are large relative to their baselines, so the main claims stand; the finer utility comparisons would need more seeds.
- **Two float-typed categoricals were modelled as continuous.** On Heart Disease, `ca` (4 values) and `thal` (3 values) are stored as floats, and the discrete-column inference used in the TVAE experiments only declared integer columns discrete. Both were therefore passed through mode-specific normalization rather than one-hot encoding, and the generator could emit non-integer values for them. The inference rule has since been corrected (bool, object and low-cardinality whole-number columns are all treated as discrete); the TVAE results in this report predate the fix. The effect is limited to two of Heart's thirteen features and applies equally to every configuration, so it does not change the comparisons, but it is a real fidelity loss on those columns.
- **No formal guarantee.** Our privacy indicators simulate specific attacks and measure their empirical success. They are not differential-privacy bounds, they do not compose, and they would not detect an attack we did not model (Stadler et al., 2022).
- **Inference risk equals downstream utility.** Because the sensitive column and the prediction target coincide in all three datasets, the inference-risk metric is numerically identical to $F1_{\text{synth}}$. It is not an independent privacy signal here. A separate sensitive attribute (for example race or sex) would make it one, and would raise the group-fairness questions we did not address.
- **Fixed-bandwidth MMD.** It saturates once the distributions separate, both as a metric and as a loss term. A multi-scale kernel would reduce this problem.
- **Absolute thresholds.** The disclosure-rate threshold (0.5) and the evaluation-time DCR are expressed in standardized raw-feature units, which grow with dimensionality, so absolute values are not comparable across datasets (baseline disclosure is 13% on Adult and 0% on Diabetes for this reason). The training margin was made relative (§6.4); the evaluation thresholds should be as well.
- **Hinge magnitude scales with dimension.** The margin m is relative but the penalty $\max(0, m - d_i)$ is not, so a fixed λ_{priv} is effectively stronger in a wider transformed space. The ordering of the knees in Table 9 is predicted by this confound as well as by the density account, so the two cannot be separated with our runs. Normalizing the hinge to $\max(0, 1 - d_i/m)$ is a one-line fix that would require re-running everything.
- **F1 discrepancy is fragile.** It is limited by noise on Heart (60-row test set) and by class imbalance on Diabetes (weighted F1 on an 89/11 target). Only on Adult was it a usable instrument. Positive-class F1 or AUROC should replace weighted F1.
- **Mean EMD is in raw units** and is dominated by high-range columns; a per-feature standardized version would allow cross-dataset comparison.
- **Fairness was out of scope, and the results show it should not have been.** The utility penalties reward faithful reproduction of the training distribution, including whatever bias it encodes, and we measured neither demographic parity nor equalized odds. The per-feature analysis (§8.6) then showed the privacy penalty shifting the sex, race and target marginals by 10 to 75 points while every aggregate utility metric improved or stayed flat. Marginal checks on protected attributes belong in the evaluation framework rather than in future work.
- **Three datasets; loss-aware penalties on two generators, one of them on one dataset only.** The density claim rests on three points on one axis, all from TVAE. The

loss-aware CTGAN experiment (§8.7) covers Heart only, at a single λ_{priv} that turned out to be past CTGAN's knee, so it establishes that the utility-term privacy gain is TVAE-specific but does not locate CTGAN's own trade-off curve. The Gaussian copula has no gradient-based training and cannot be made loss-aware in this sense.

- **Baseline training budgets were library defaults.** CTGAN's Heart result is an artefact of `batch_size=500` on 237 rows, not a property of the architecture. A fair comparison would tune the budgets per dataset as we did for the loss-aware runs.

11 Conclusion

We started this project with the goal of building a synthetic-data generator that minimizes utility loss under an explicit privacy constraint, and with the argument that writing the constraint into the objective is ethically preferable to checking it afterwards. We built the generator and we still hold the argument. The experiments, however, returned something we had not asked for. The privacy penalty protects individuals by moving generated records into the least populated regions of every attribute. As a consequence it inflates minority classes and rare categories and misstates the outcome rate, and it does so more strongly the sparser the data. Our evaluation framework, eight metrics that are standard in the literature, reported an improvement on the dataset where the sex ratio had moved by fourteen points. That the populations for whom privacy matters most are the ones for whom it is most expensive, and that the instruments meant to detect the expense cannot see it, is an uncomfortable pair of results. It is also the kind of result an ethics-first framing is supposed to bring out. The loss-aware generator does not make the decision easier. It makes it visible, per dataset, as four numbers, and it gives whoever sets those numbers the evidence that they must be checked column by column, because the aggregate metrics will not show what was paid.

Code and data. All code, run reports, figures and the extended technical documentation are at `https://github.com/miky-alt/Loss-Aware-Synthetic-Data-Generation`. Every number in this report can be reproduced from the run index with `src/experiments/plot_tradeoff.py` and `src/experiments/plot_feature_emd.py`.

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A The loss-aware training loop

Algorithm 1 is the body of `LossAwareTVAE.fit()`. Lines 1–3 and the block from line 9 onward are unchanged from `ctgan.TVAE`; lines 4–5 and 12–20 are our additions. With $\lambda_{\text{mmd}} = \lambda_{\text{corr}} = \lambda_{\text{priv}} = 0$ the inserted block is skipped entirely.

Algorithm 1 Loss-aware TVAE training

Require: training table D; discrete columns; weights $\lambda_{\text{mmd}}, \lambda_{\text{corr}}, \lambda_{\text{priv}}$; relative margin μ; bandwidth γ

1: $T \leftarrow \textsc{DataTransformer}(D)$; $X \leftarrow T(D)$ ▷ mode-specific normalization + one-hot
2: initialise encoder E_ϕ, decoder G_θ, optimiser (Adam, weight decay ℓ_2)
3: $\tilde{d}_{\text{real}} \leftarrow \text{median}_i \min_{j \neq i} \|X_i - X_j\|_2$ ▷ on ≤ 2000 rows
4: $m \leftarrow \mu \cdot \tilde{d}_{\text{real}}$ ▷ effective privacy margin
5: **for** epoch $= 1 \dots E$ **do**
6:     **for** each minibatch $x \in X$ of size B **do**
7:         $\mu_z, \sigma_z, \log \sigma_z^2 \leftarrow E_\phi(x)$; $z \leftarrow \mu_z + \sigma_z \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$
8:         $\hat{x}, \sigma_{\text{dec}} \leftarrow G_\theta(z)$
9:         $\mathcal{L} \leftarrow \mathcal{L}_{\text{recon}}(\hat{x}, x, \sigma_{\text{dec}}) + \mathcal{L}_{\text{KL}}(\mu_z, \log \sigma_z^2)$ ▷ stock TVAE
10:         **if** $\lambda_{\text{mmd}} > 0$ or $\lambda_{\text{corr}} > 0$ or $\lambda_{\text{priv}} > 0$ **then**
11:             $z' \sim \mathcal{N}(0, I)^B$; $r \leftarrow G_\theta(z')$ ▷ fresh prior batch, not a reconstruction
12:             **if** $\lambda_{\text{mmd}} > 0$ or $\lambda_{\text{corr}} > 0$ **then**
13:                 $\tilde{x} \leftarrow \sigma_{\text{soft}}(r)$ ▷ tanh on scalars, softmax on one-hot spans
14:                 $\mathcal{L} \leftarrow \mathcal{L} + \lambda_{\text{mmd}} \cdot \text{MMD}_\gamma^2(\tilde{x}, x) + \lambda_{\text{corr}} \cdot \|\text{Corr}(\tilde{x}) - \text{Corr}(x)\|_F$
15:             **end if**
16:             **if** $\lambda_{\text{priv}} > 0$ **then**
17:                 $\tilde{x}^{\text{hard}} \leftarrow \sigma_{\text{hard}}(r)$ ▷ straight-through one-hot on one-hot spans
18:                 $d_i \leftarrow \min_j \|\tilde{x}_i^{\text{hard}} - x_j\|_2$ for $i = 1..B$
19:                 $\mathcal{L} \leftarrow \mathcal{L} + \lambda_{\text{priv}} \cdot \frac{1}{B} \sum_i \max(0, m - d_i)$
20:             **end if**
21:         **end if**
22:         **if** $\mathcal{L}$ is not finite **then skip step**
23:         **end if**
24:         backpropagate $\mathcal{L}$; optimiser step; clamp $\sigma_{\text{dec}} \in [0.01, 1]$
25:         log $(\mathcal{L}_{\text{recon}}, \mathcal{L}_{\text{KL}}, \mathcal{L}_{\text{mmd}}, \mathcal{L}_{\text{corr}}, \mathcal{L}_{\text{priv}}, \bar{d})$
26:     **end for**
27: **end for**

Sampling is unchanged from TVAE: draw $z \sim \mathcal{N}(0, I)$, decode, apply tanh and arg max, and invert the transformer.

Algorithm 2 is the corresponding loop for `LossAwareCTGAN.fit()` (§6.3). The discriminator steps are unchanged from `ctgan.CTGAN`; the inserted block (lines 11–16) mirrors Algorithm 1's penalty block applied to the generator's own fake batch rather than a fresh prior sample, since CTGAN already produces one every step.

B Full per-feature EMD tables

Tables 10 and 11 report every column for Heart Disease and Adult. The Diabetes table has 45 columns and is provided as `experiments/figures/feature_emd_diabetes_batch_size500.csv`. All values are mean EMD over seeds, in raw feature units. k is the number of distinct values in the real column.

Algorithm 2 Loss-aware CTGAN training (generator step only; discriminator steps unchanged from stock CTGAN)

Require: training table D; discrete columns; weights $\lambda_{\text{mmd}}, \lambda_{\text{corr}}, \lambda_{\text{priv}}$; relative margin μ; bandwidth γ

$T \leftarrow \textsc{DataTransformer}(D)$; $\quad X \leftarrow T(D)$; initialise G_θ, D_ψ, optimisers
$\tilde{d}_{\text{real}} \leftarrow \text{median}_i \min_{j \neq i} \|X_i - X_j\|_2$; $\quad m \leftarrow \mu \cdot \tilde{d}_{\text{real}}$
for epoch $= 1 \ldots E$ **do**
  for each step **do**
    ▷ discriminator steps: unchanged from stock CTGAN, omitted here
    $c \leftarrow$ sampled condition vector; $z \sim \mathcal{N}(0, I)$
    $\tilde{x}_{\text{raw}} \leftarrow G_\theta(z, c)$; $\quad \tilde{x} \leftarrow \sigma(\tilde{x}_{\text{raw}})$ ▷ tanh / softmax, as in Alg. 1
    $\mathcal{L}_G \leftarrow -D_\psi(\tilde{x}, c) + \text{CE}(\hat{c}, c)$ ▷ stock CTGAN generator loss
    if $\lambda_{\text{mmd}} > 0$ or $\lambda_{\text{corr}} > 0$ or $\lambda_{\text{priv}} > 0$ **then**
      $x \leftarrow$ real batch sampled under condition c
      if $\lambda_{\text{mmd}} > 0$ or $\lambda_{\text{corr}} > 0$ **then**
        $\mathcal{L}_G \leftarrow \mathcal{L}_G + \lambda_{\text{mmd}} \cdot \text{MMD}^2_\gamma(\tilde{x}, x) + \lambda_{\text{corr}} \cdot \|\text{Corr}(\tilde{x}) - \text{Corr}(x)\|_F$
      end if
      if $\lambda_{\text{priv}} > 0$ **then**
        $\tilde{x}^{\text{hard}} \leftarrow \sigma_{\text{hard}}(\tilde{x}_{\text{raw}})$; $\quad d_i \leftarrow \min_j \|\tilde{x}^{\text{hard}}_i - x_j\|_2$
        $\mathcal{L}_G \leftarrow \mathcal{L}_G + \lambda_{\text{priv}} \cdot \frac{1}{B} \sum_i \max(0, m - d_i)$
      end if
    end if
    if $\mathcal{L}_G$ is not finite **then skip step**
    end if
    backpropagate $\mathcal{L}_G$; generator optimiser step
    log $(\mathcal{L}_G, \mathcal{L}_D, \mathcal{L}_{\text{mmd}}, \mathcal{L}_{\text{corr}}, \mathcal{L}_{\text{priv}}, \bar{d})$
  end for
end for

Table 10: Heart Disease: per-feature EMD by λ_{priv} ($\mu = 1.5$; baseline has all $\lambda = 0$). For $k = 2$ columns the value is the absolute difference in proportion.

column	k	baseline	λ_{priv}=1	λ_{priv}=2	λ_{priv}=5	λ_{priv}=10	λ_{priv}=50
sex	2	0.057	0.043	0.044	0.041	0.118	0.209
fbs	2	0.122	0.046	0.050	0.062	0.573	0.092
exang	2	0.059	0.068	0.054	0.062	0.241	0.125
num (target)	2	0.013	0.038	0.037	0.052	0.229	0.187
restecg	3	0.264	0.119	0.141	0.107	0.278	0.362
slope	3	0.116	0.095	0.075	0.091	0.184	0.354
thal	3	0.405	0.302	0.298	0.311	0.932	1.391
cp	4	0.197	0.157	0.171	0.181	0.402	1.505
ca	4	0.276	0.188	0.205	0.200	0.224	0.400
oldpeak	40	0.135	0.221	0.259	0.248	0.573	1.490
age	41	2.02	1.87	2.00	2.06	2.82	20.6
trestbps	50	5.28	3.45	3.25	4.35	8.06	58.7
thalach	91	3.70	2.87	3.30	3.63	4.96	61.7
chol	152	6.33	7.53	9.33	9.13	14.8	280.5

Table 11: UCI Adult: per-feature EMD, baseline vs $\lambda_{\text{priv}} = 10$ ($\mu = 1.5$), with the ratio and the standardized change used for the ranking in §8.6. Sorted by standardized change.

column	k	baseline	λ_{priv}=10	ratio	std. change
native-country	42	1.447	9.085	6.28	+1.04
race	5	0.156	0.913	5.85	+0.90
workclass	9	0.131	0.556	4.26	+0.31
sex	2	0.010	0.144	13.81	+0.28
income (target)	2	0.013	0.111	8.26	+0.23
relationship	6	0.106	0.465	4.39	+0.22
age	74	1.583	3.339	2.11	+0.13
capital-gain	122	414.6	1266.2	3.05	+0.11
marital-status	7	0.127	0.227	1.78	+0.07
occupation	15	0.494	0.705	1.43	+0.05
capital-loss	98	62.8	71.7	1.14	+0.02
education-num	16	0.451	0.356	0.79	−0.04
hours-per-week	96	3.142	2.012	0.64	−0.09
education	16	1.125	0.367	0.33	−0.20
fnlwgt	27946	65810	38617	0.59	−0.26

C Reproducing every number

Every run writes a JSON report to `experiments/results/` and a line to `experiments/results/index.jsonl`. The two plotting scripts read the index, keep the most recent run for each (configuration, seed) pair, aggregate across seeds and write the CSV files from which every table above was taken.

Runs. With `H="-dataset heart -generator tvae_loss_aware -kwarg batch_size=32 -kwarg epochs=300"` and `U="-kwarg lambda_mmd=1 -kwarg lambda_corr=0.5"`:

- Heart ablation (Table 3): `run $H -seed s` with no kwargs (baseline), `$U` (utility only), `-kwarg lambda_priv=1 -kwarg dcr_margin=1` (privacy only), both; $s \in \{1, 2, 3\}$.
- Heart sweep (Table 4): `run $H $U -kwarg lambda_priv=L -kwarg dcr_margin=M -seed s` for $(L, M) \in \{1, 2, 5, 10\} \times \{1.5\}$ with $s \in 1..5$ for $L \in \{2, 5, 10\}$; and $(L, M) \in \{10, 50\} \times \{1.5, 2.0\}$, $s \in 1..3$.
- Adult (Table 5) and Diabetes (Table 6): as above with `-kwarg batch_size=500 -kwarg epochs=100`; $L \in \{0, 2, 5, 10\}$ for Diabetes, $\{0, 10\}$ for Adult; $s \in 1..3$.
- Baselines (Table 2): `run -dataset d -generator g -seed s` for $g \in \{$`ctgan`, `tvae`, `gaussian_copula`$\}$, library defaults.

Figures and tables.

- `python -m src.experiments.plot_tradeoff -dataset heart -kwarg batch_size=32 -kwarg epochs=300 -exclude-collapse 4` (Figure 1); the same for `adult` and `diabetes` with `-kwarg batch_size=500`.
- `python -m src.experiments.plot_feature_emd -dataset D ...` with the same filters (Figures 4 and 5, Tables 7, 10, 11).

Environment. Python 3.12, `torch` 2.13, `ctgan` 0.11.1, `sdv` 1.32.1, `scikit-learn` 1.9, on Apple silicon (MPS). Seeds control the train/test split, model initialization and sampling. MPS kernels are not bitwise deterministic, so re-running reproduces the tables up to seed noise rather than exactly.

D Original project proposal

We reproduce the abstract and the deliverables from our initial project proposal, submitted before any experiments were run, so that the reader can compare what was planned with what was found. §1.1 discusses the differences.

Abstract (original). The generation of synthetic data presents a fundamental trade-off between preserving individual privacy and retaining the statistical utility of the original dataset. This project investigates methodologies to measure and mitigate the utility loss inherently introduced during the generation process through loss-aware optimization. We define utility loss as the quantitative gap between real and synthetic data concerning distributional fidelity, feature correlation preservation, and downstream predictive performance (evaluated via metrics such as MMD, EMD, and F1-score discrepancy). Concurrently, privacy constraints are strictly monitored using established indicators including DCR, Overfitting Protection, Inference Risk, CM3, and Disclosure Protection. To transition from mere evaluation to active prevention, this research explores the integration of explicit penalty terms within the training objectives of synthetic data generators. By designing a multi-objective loss function that penalizes distribution mismatch, correlation distortion, and privacy leakage during the training phase, we aim to systematically balance utility and privacy. The ultimate goal is to deliver a computational framework capable of estimating utility loss, evaluating generator architectures, and actively minimizing information degradation.

Methodology and goals (original). The central vision of this project is to shift the paradigm of synthetic data generation from post-generation evaluation to active, in-training optimization. The first phase focuses on formally defining and estimating utility loss, articulated as the gap between real and synthetic data across distributional fidelity, feature correlation preservation, and predictive performance, quantified with MMD, EMD and downstream F1-score discrepancy, while privacy is monitored with DCR, Inference Risk, CM3 and explicit Disclosure Protection constraints. The second phase transitions from measurement to prevention, modifying the core training objective of synthetic data generators by integrating explicit penalty terms for distribution mismatch, correlation distortion, downstream performance degradation, and privacy leakage.

Expected deliverables (original).

1. **A Computational Framework:** a robust toolset for accurately estimating the utility loss introduced by synthetic data, factoring in both statistical properties and downstream application performance.
2. **Architecture Analysis:** the identification and comparative analysis of synthetic data generator architectures to determine which inherently minimize information loss.
3. **Loss Function Design:** the mathematical formulation, implementation, and empirical validation of a novel multi-objective loss function designed to actively reduce utility loss while preserving stringent privacy guarantees during training.